

Python Project

Detecting & Measuring Bias in AI Based Decision Models

By: Loïs Kindomba, Selena Cha, Tejpreet Jaswal

Agenda

- Business Understanding
- Data Understanding
- Dataset Structure, Features, and Domain Attributes
- Data Preparation
- Modeling
- Results and Evaluation
- Findings and Recommendations
- Faith and Ethics Implications

Business Understanding

- AI is widely used for toxicity detection
 - Helps platforms manage harmful content
 - Language complexity makes moderation challenging
- Risk of unintended model bias
 - Example: Perspective API mislabeling neutral identity statements
 - Research shows identity terms often increase toxicity scores unfairly
- Why this matters
 - Can silence certain demographic groups
 - Affects platform trust, safety, and ethical responsibilities
- Project focus
 - Evaluate how models behave
 - Identify signs of bias linked to identity references





DATA UNDERSTANDING & PREPARATION

Data Understanding

Dataset: Jigsaw Unintended Bias in Toxicity Classification

- ~450,000 online comments
- Continuous toxicity score (0–1)

Additional toxicity indicators

- Insult, threat, identity attack
- Severe toxicity, obscene, sexual content

Identity attributes

- Gender, race/ethnicity, religion
- Sexual orientation, disability references

Purpose

- Supports both toxicity prediction and fairness analysis

Dataset Structure, Features, and Domain Attributes

Dataset Description and Variables

The dataset includes a rich mix of feature groups:

- **Text Feature:** comment_text
- **Toxicity Labels:** severe_toxicity, obscene, insult, identity_attack, threat, sexual_explicit
- **Identity Attributes:** gender (male, female, transgender), race/ethnicity (white, black, asian, latino), religion (christian, muslim, jewish, hindu), disabilities, and sexual orientation indicators
- **Engagement Metadata:** rating, likes, funny, wow, sad, disagree
- **Annotator Information:** identity_annotator_count, toxicity_annotator_count
- **Identifiers/Dates:** id, article_id, publication_id, created_date

This combination supports both toxicity prediction and fairness analysis by highlighting how comments reference specific identity groups. The data was loaded into a pandas dataframe (df) with 45 columns and hundreds of thousands of rows.

Data Preparation – Target Variable

- **Original toxicity score**
 - Continuous range from 0 to 1
- **Converted to binary label**
 - Threshold at 0.5
 - toxic vs. non-toxic
- **Why**
 - Aligns with real moderation decisions
 - Enables consistent model comparison



Data Preparation – Features



Text preprocessing

Cleaning and standardizing
Lowercasing and tokenizing



TF-IDF vectorization

Converts text into numerical features
Handles punctuation removal



Additional features included

Identity attributes
Metadata and annotator information

Data Quality Considerations

- **Missing identity attributes**
 - Due to comments not mentioning groups
 - Filled missing values with zero
- **Class imbalance**
 - Majority of comments are non-toxic
 - No oversampling or class-weighting applied
- **Impact**
 - Imbalance influences recall
 - Models trained under natural distribution for fair comparison





MODELING

Overall Modeling Approach

Three classification models were trained to evaluate their performance on toxicity prediction: **Logistic Regression**, **Random Forest**, and **XGBoost**. This selection provides a balanced comparison between linear methods, tree-based ensembles, and advanced boosting techniques. All models were trained on identical TF-IDF text features and identity attributes to maintain fairness. Logistic Regression served as a baseline linear model, Random Forest captured nonlinear interactions, and XGBoost known for state-of-the-art result provided a more expressive model capable of handling complex relationships in the data.

Model Specifications and Parameters

Model Configurations and Rationale

- **Logistic Regression** used max_iter=1000 to ensure convergence on the high-dimensional TF-IDF data, providing an interpretable baseline.
- **Random Forest** was used with default settings, leveraging hundreds of unpruned trees to capture diverse text–identity interactions without heavy tuning.
- **XGBoost** used tuned parameters (300 trees, depth 6, 0.8 subsampling) to reduce overfitting and better model subtle patterns in toxicity.

These configurations allowed each model to represent a distinct modeling philosophy from interpretable linearity, to flexible ensembles, to powerful gradient boosting providing strong comparative insight into toxicity detection performance.

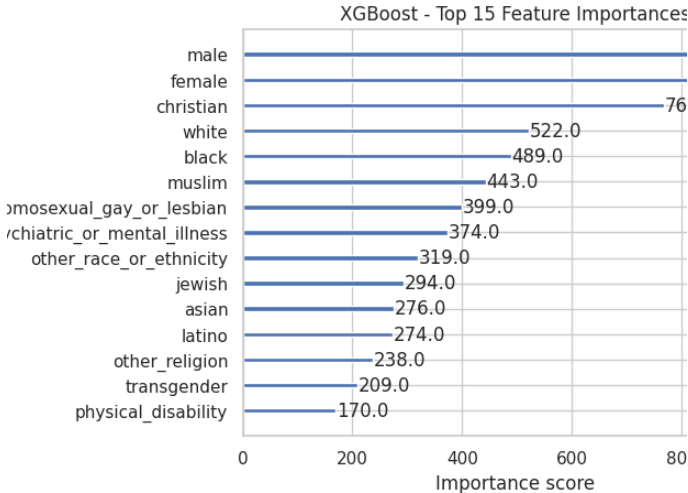
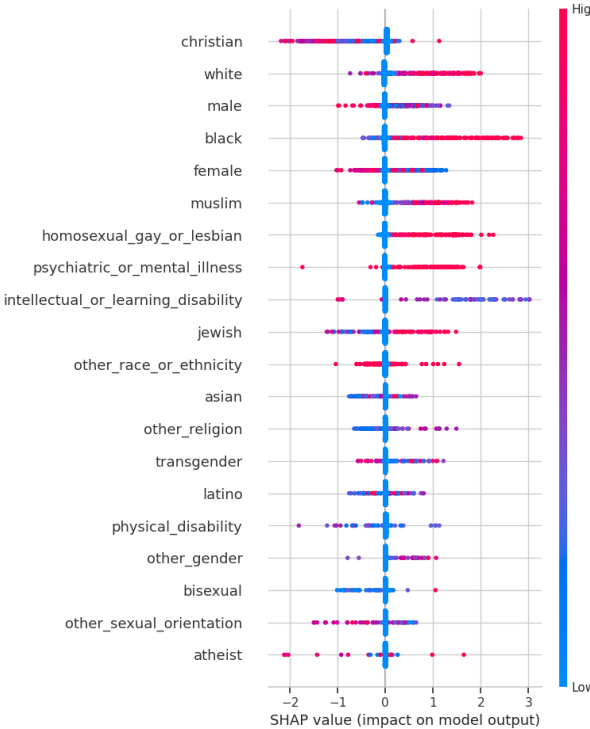
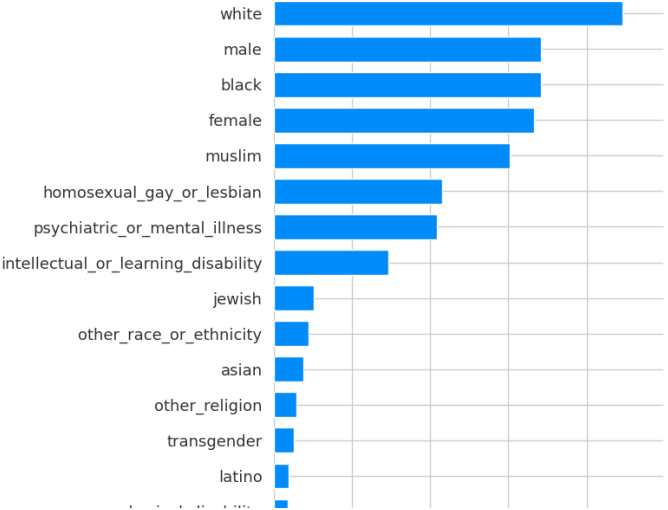
Explainability Through SHAP Analysis

Model Interpretability and Bias Assessment

SHAP values were used to explain predictions from the XGBoost model. SHAP summary plots (both bar and beeswarm) revealed that text vocabulary was by far the strongest driver of predictions, while identity attributes had extremely low SHAP influence.

This indicates that the model relied on language cues rather than sensitive demographic references, reducing the risk of identity-based bias. SHAP analysis therefore allowed transparency into the model's decision process and supported an ethical evaluation of the classifier's behavior.

Shap analysis and Important Features



RESULTS & EVALUATION

Model Performance Metrics

Accuracy Results and Classification Outputs

All three models produced similar accuracy:

- Logistic Regression: **0.5441**
- Random Forest: **0.5449**
- XGBoost: **0.5453**

Classification reports showed identical patterns:

- **Non-toxic class (0):** very high recall (~0.94) and strong F1 (~0.67)
- **Toxic class (1):** low recall (~0.15) and low F1 (~0.24–0.25)

These results reflect the severe class imbalance—models overwhelmingly favor predicting “non-toxic,” missing many harmful comments. Accuracy alone is misleading, making precision, recall, and F1-score essential for evaluation.

Comparative Insights and Model Ranking

Model Comparison and Strength Assessment

XGBoost was the strongest performer overall, achieving the highest accuracy and better handling of nonlinear text patterns compared to Logistic Regression and Random Forest. Logistic Regression remained the most interpretable model but performed the weakest on nonlinear interactions. Random Forest provided moderate improvement but struggled with the high dimensionality of TF-IDF features.

Across criteria predictive power, robustness to imbalance, and ability to capture subtle language features **XGBoost ranked best**, though all models shared difficulty identifying toxic comments due to data imbalance.

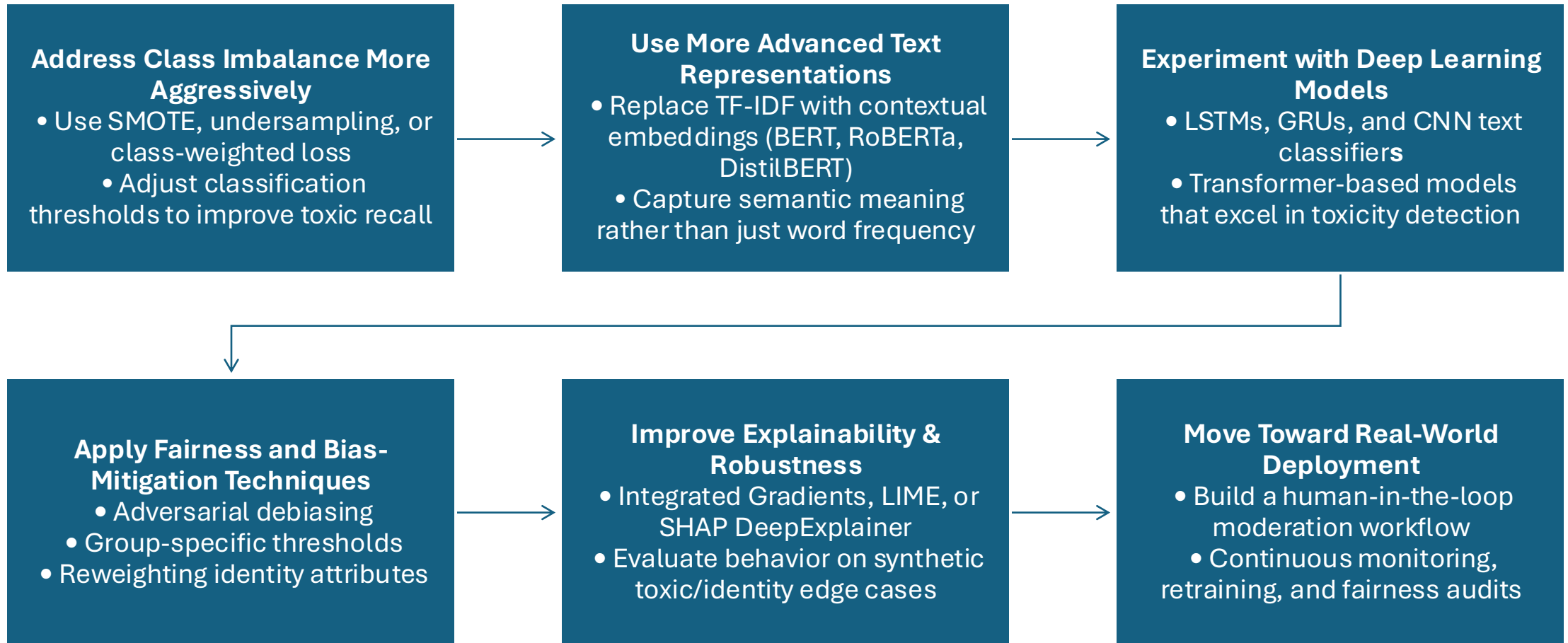
Ethical Interpretation, SHAP Findings, and Practical Implications

Final Evaluation and Real-World Considerations

The models performed well at identifying non-toxic comments but struggled significantly with detecting truly toxic language, leading to low recall for the toxic class. This poses serious risks in safety-critical applications, as harmful content may go unflagged. SHAP analysis confirmed that predictions were primarily driven by text patterns rather than identity features, indicating no major evidence of identity-based bias.

However, given the low F1-scores for toxic comments, none of the models are suitable for automated moderation on their own. They should instead act as decision-support tools alongside human reviewers. Additional techniques resampling, threshold adjustments, or contextual embeddings are needed for production-level fairness and accuracy.

Future Enhancements and Advanced Modeling Options





Findings & Recommendations

Findings

- Models showed high overall accuracy but consistently low toxic recall, meaning many harmful comments were not detected.
- Non-toxic predictions dominated, revealing a strong bias toward the majority class caused by data imbalance.
- SHAP analysis showed identity-related terms had a significant influence on predictions, creating fairness and consistency risks.
- Missed toxic comments and identity-sensitive behavior indicate that the model is not reliable or ethically safe for deployment.
- These findings show clear needs for fairness audits, improved toxic data representation, stronger documentation, human oversight, and responsible AI training

Recommendation 1: Fairness & Performance Audit

(Highest Priority)

- Needed because models showed low toxic recall despite high accuracy
- Reveals hidden issues: identity-term influence and uneven class performance
- Tracks metrics by class, subgroup, and identity context
- Identifies bias patterns early and prevents harmful deployment errors
- Includes recurring audit reports, SHAP monitoring, and performance dashboards
- Improves accountability, reliability, and ethical transparency

Decision

Implement first because all other improvements depend on accurate, ongoing measurement

Challenge: ongoing workload and technical capacity

Recommendation 2: Improve Data Quality & Representation

- Toxic class imbalance caused systematic model failure on harmful comments
- Expanding toxic samples captures broader forms of harassment and subtle toxicity
- Balancing the dataset reduces the model's dependence on identity terms
- Enhances fairness by helping the model distinguish identity mention vs identity attack
- Uses class weighting, curated toxic data, and review of identity-related features
- Results in higher toxic recall, stronger generalization, and safer outputs

Decision

Prioritize because data imbalance is the main cause of model failure

Challenge : Time-consuming annotation and careful curation needed

Recommendation 3: Strengthen Transparency & Documentation

- Responds to identity-term influence revealed in SHAP results
- Documentation clarifies how preprocessing, TF-IDF, feature engineering, and model choices shape fairness outcomes
- Supports internal review, ethical evaluation, and reproducibility
- Helps diagnose unexpected behaviors and prevents accidental fairness regressions
- Makes model decisions traceable and reviewable
- Builds trust and allows technical teams to justify moderation decisions

Decision: adopt because transparency is essential for diagnosing fairness issues

Challenge: requires consistent maintenance

Recommendation 4: Integrate Human Oversight

- Addresses frequent toxic false negatives
- Adds human review for low-confidence and identity-sensitive cases
- Provides context and nuance the model cannot capture
- Essential for borderline cases and comments involving identity terms
- Reduces harmful misclassifications and prevents biased automated decisions
- Strengthens fairness and ensures protection for vulnerable groups

Decision

Integrate selective oversight until toxic recall and identity-handling until model reliability improves

Challenge: slower moderation and increased labor

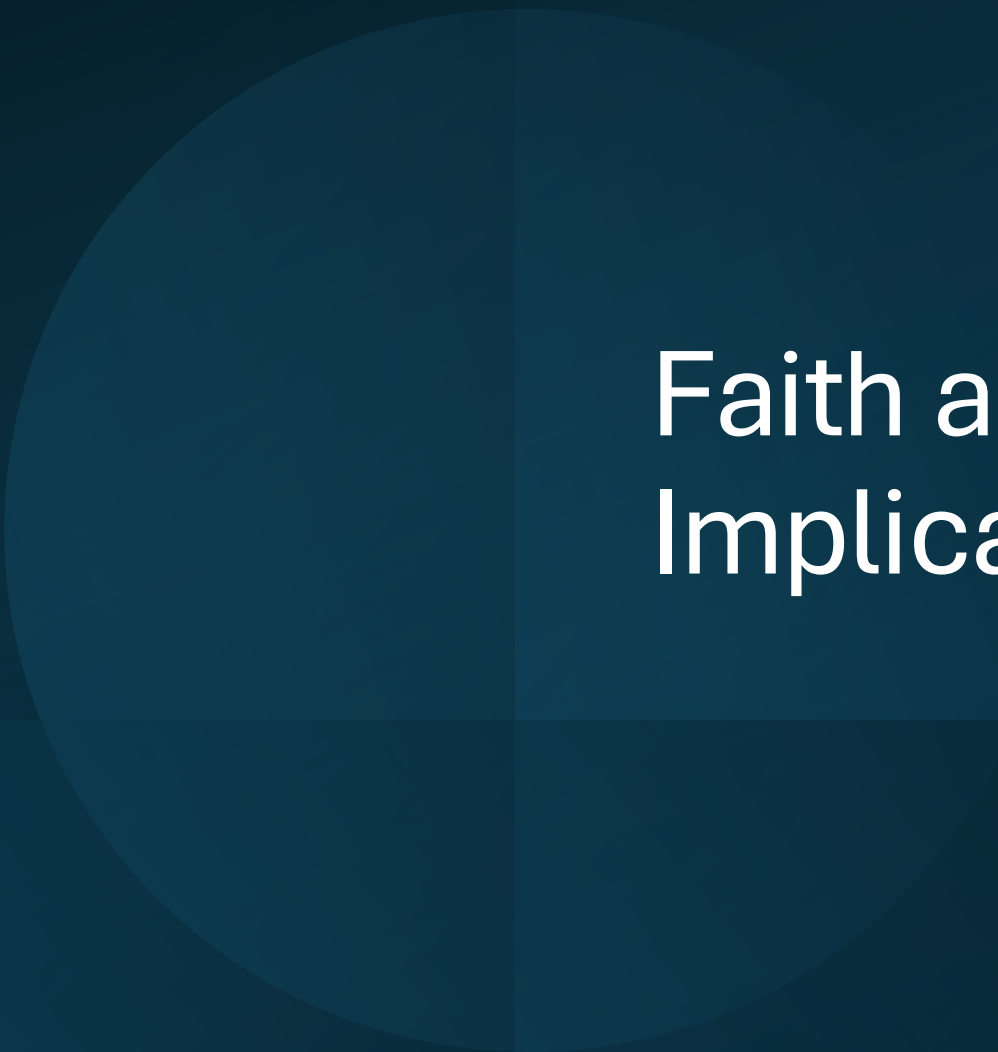
Recommendation 5: Training for Responsible AI

- Ensures teams understand fairness metrics and bias issues
- Supports correct handling of identity-related content
- Builds long-term capacity to manage ethical risks and make consistent decisions
- Reinforces fairness as an organizational practice, not a one-time fix
- Includes workshops on fairness auditing, data ethics, and responsible moderation
- Supports correct use of documentation, audits, and hybrid workflows

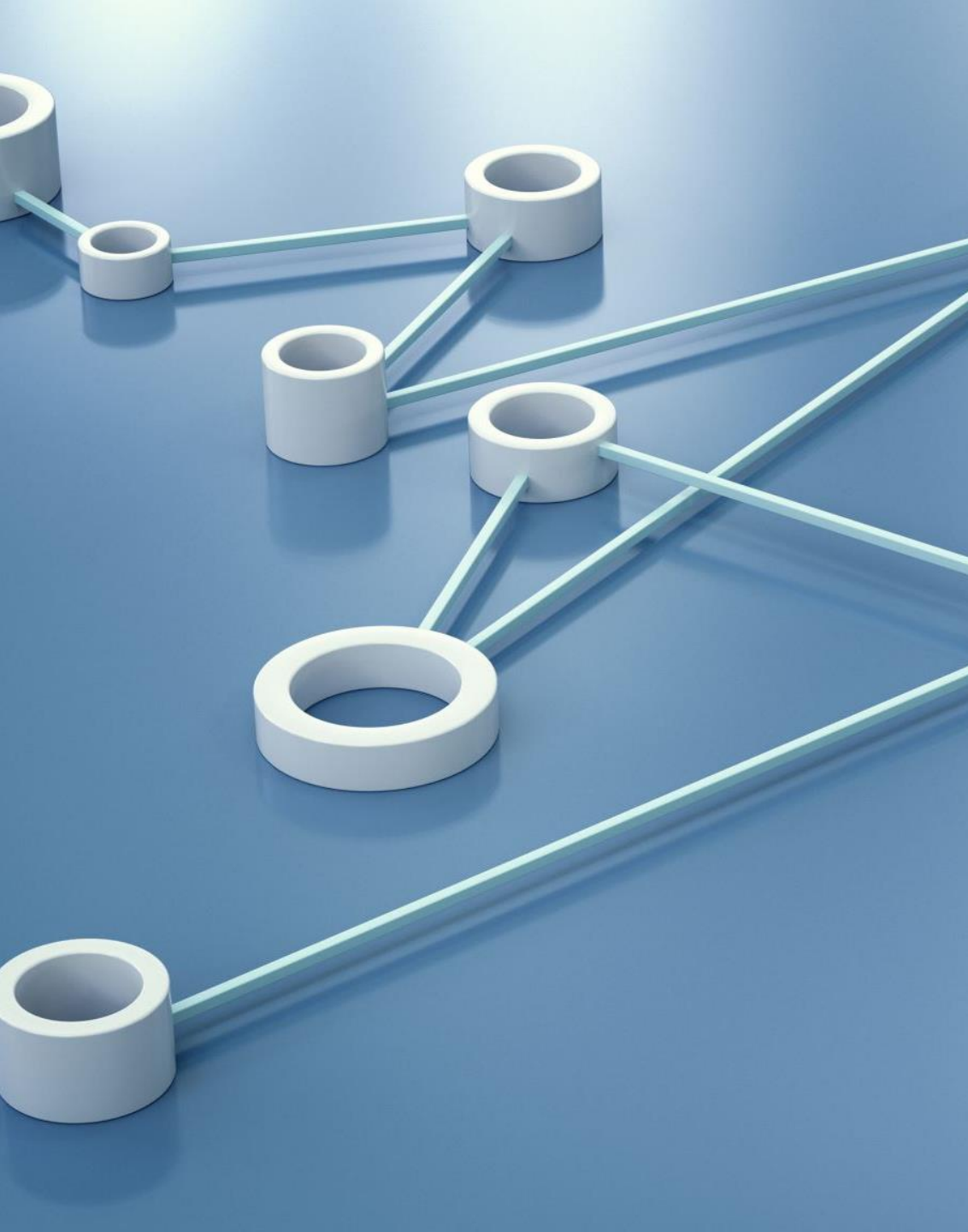
Decision

Implement to sustain all fairness improvements over time

Challenge: requires time and resources



Faith and Ethics Implications



Ethical Implications of the Project

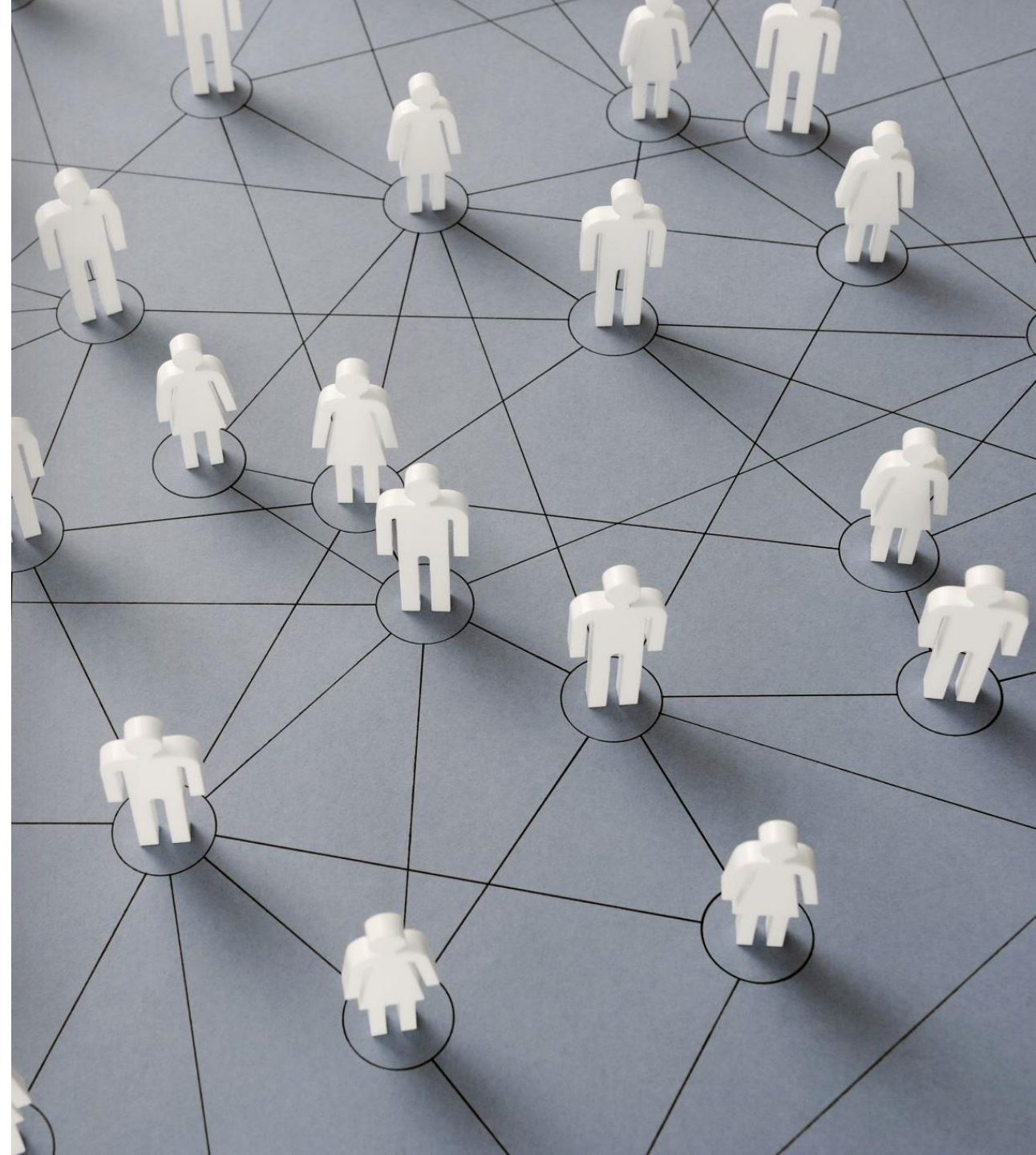
- Low toxic recall allows harmful content to remain unflagged, creating safety risks for users, especially vulnerable groups.
- Identity term influence raises fairness concerns because the model may treat identity-related language inconsistently.
- Accuracy metrics alone are not enough, since error distribution reveals ethical and real-world consequences.
- Automated decisions without oversight can amplify harm when bias or imbalance is present.
- Ethical responsibility requires transparent evaluation, ongoing monitoring, and correction of model weaknesses.

Christian Faith Perspective & Connection

- Christian values call for fairness, justice, and protection of individuals who may be harmed or overlooked.
- Impartial judgment aligns with the need to ensure AI systems do not unfairly target or neglect specific groups.
- Stewardship emphasizes responsible development and use of technology that affects people's well-being.
- Respect for human dignity supports fairness checks, transparency practices, and careful treatment of identity-related content.
- The faith perspective reinforces the call to use data and analytics in ways that promote equity, safety, and compassion.

Broader Ethical and Faith Implications

- The project illustrates how AI systems can influence online interactions, public discourse, and community safety.
- Bias or imbalance in these systems can scale into larger societal inequities if not corrected.
- Ethical AI practices benefit organizations, users, and society by improving trust, safety, and fairness.
- From a faith perspective, building responsible AI echoes broader commitments to justice, accountability, and care for others.
- The implications extend beyond this project, demonstrating the ongoing need for ethical thinking in all data-driven decisions.



Conclusion

- The project showed that improving toxicity detection requires strong technical methods and ethical responsibility.
- Findings confirmed gaps in toxic recall and identity handling that must be addressed for fair and safe AI.
- Our recommendations provide a practical path toward better accuracy, transparency, and oversight.
- The faith and ethics perspective emphasized fairness, dignity, and accountability in AI decisions.
- Overall, trustworthy AI requires continuous evaluation and a commitment to protecting all users.

The background of the slide features three overlapping circles in a teal color, positioned horizontally across the top and bottom. A white horizontal band runs through the center of the image, containing the text.

Thank You!